# Introduction

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# 1 Motivation and formalization

In order to obtain an estimate of an unknown quantity, say $\mu_0$, for example a speed, it is common to take $n$ measurements $x_1, \ldots, x_n$ and calculate their mean:

$$\bar{x}_n = \frac{x_1 + x_2 + \ldots + x_n}{n}.$$

But why should the observations be combined in this way ?

$\rightarrow$ the empirical mean is a relevant measure of the center of the observations, since

$$\bar{x}_n = \arg\min_a \sum_{i=1}^{n} (x_i - a)^2.$$

But (at this level) we can't justify why $\bar{x}_n$ is a good estimate (approximation) of the true value $\mu_0$ since no explicit assumption has been made to connect the data $(x_1, \ldots, x_n)$ and $\mu_0$.

To establish such a connection, we can, for example, presume that:

(i) each $x_i$ is an observed value of a rv $X_i$, and
(ii) $X_i$, $i = 1, \ldots, n$, have a common mean $\mu_0$.

Even more specifically, we can, for example, assume the following ***additive error model***

$$X_i = \mu_0 + \epsilon_i,\ \epsilon_i \sim N(0, \sigma_0^2),$$

or equivalently $X_i \sim N(\mu_0, \sigma_0^2)$. In this way, the problem we face is the estimation of $\mu_0 = E(X_i)$ from the sample $(X_1, \ldots, X_n)$.

## 1.1 General formalization

We observe a realization (data) $\boldsymbol{x} := (x_1, \ldots, x_n)^t$ of a rve (sample) $\boldsymbol{X} := (X_1, \ldots, X_n)^t$ taking values in a sample space $\boldsymbol{\mathcal{X}} \subset \mathbb{R}^n$ according to a (partially) *unknown joint distribution* $\boldsymbol{P}$ conveniently specified by a pd.

**Definition 1.1.** A statistical model is a collection of assumptions about $\boldsymbol{P}$, the underlying distribution of the observed sample.

$X_1, \ldots, X_n$ are typically assumed to be ***iid*** copies of some population rv, which we denote by $X$. The ***common distribution***, i.e. the distribution of the $X_i$'s, is denoted by $P$. In this case the statistical model reduces to a collection

of assumptions about $P$. We write such a model as $P \in \mathcal{P}$, where $\mathcal{P}$ is a ***given family*** of probability measures satisfying certain conditions.

### The IID hypothesis plays a crucial role

Without the iid assumption the statistical analysis of the data becomes much more complicated. For example, as a statistical model, we could assume that

$$\boldsymbol{X} \sim N_n(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0),$$

with an unknown $\boldsymbol{\theta}_0 := (\mu_1, \ldots, \mu_n, \sigma_{11}, \ldots, \sigma_{nn})^t \in \mathbb{R}^{n+\frac{n\times(n+1)}{2}}$, where $\mu_i = E(X_i)$ and $\sigma_{ij} = Cov(X_i, X_j)$.

Now, under the iid assumption, $\boldsymbol{\theta}_0$ reduces to $(\mu_1, \sigma_{11})^t \in \mathbb{R}^2$.

In general, under the iid assumption, if the pd of $X$ is $f$ and its cdf is $F$, then the joint pd and the joint cdf of $\boldsymbol{X}$, denoted by $f_n$ and $F_n$, respectively, are given by

$$f_n(\boldsymbol{x}) := f_n(x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i) \text{ and } F_n(\boldsymbol{x}) := P(X_1 \leq x_1, \ldots, X_n \leq x_n) = \prod_{i=1}^{n} F(x_i).$$

## 1.2 Parametric models

For a parametric model, the distribution $P$ of $X_i$ is completely determined by a ***finite*** (fixed) number of parameters, i.e.

$$P \in \mathcal{P} = \{P_{\boldsymbol{\theta}},\ \boldsymbol{\theta} \in \Theta\},$$

where $\Theta \subset \mathbb{R}^d$, for some fixed positive integer $d$. $\Theta$ is called ***the parameter space***. Each $P_{\boldsymbol{\theta}}$ is a completely known pd when the *generic parameter* $\boldsymbol{\theta}$ is known. The true value of $\boldsymbol{\theta}$ is typically denoted by $\boldsymbol{\theta}_0$.

**Example 1.1.**

- The Bernoulli model:
$$\mathcal{P} = \left\{ f(x;\theta) = \theta^x (1-\theta)^{(1-x)} I(x \in \{0,1\}),\ \theta \in [0,1] \right\},$$
i.e. $X \sim Ber(\theta_0)$, for a particular but unknown $\theta_0 \in [0,1] = \Theta$.

- The Exponential model:
$$\mathcal{P} = \left\{ f(x;\theta) = \theta^{-1} e^{-x/\theta} I(x > 0),\ \theta > 0 \right\},$$
i.e. $X \sim Exp(\theta_0)$, for some unknown $\theta_0 \in (0,\infty) = \Theta$.

- The Normal model:

$$\mathcal{P} = \left\{ f(x;\boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} I(x \in \mathbb{R}),\ \boldsymbol{\theta} = (\mu, \sigma^2)^t \in \mathbb{R} \times (0, \infty) \right\},$$

  i.e. $X \sim N(\mu_0, \sigma_0^2)$, for some unknown $\mu_0$ and $\sigma_0^2$. In this case $\Theta = \mathbb{R} \times (0, \infty)$. $\square$

If the distribution $P$ of $X_i$ is not completely determined by a finite number of parameters, then the model is *semiparametric* or *nonparametric*.

**Example 1.2.**

- $X$ has a pd $f$, with $\int f''(x)dx < \infty$ and/or $\int x^2 f(x)dx < \infty$.
- $X$ has a symmetric distribution (pd) around 0, i.e. $F(-x) = 1 - F(x)$.
- $X$ has a cdf $F$ satisfying $F(x) = F_0(x - \mu)$ for some $\mu \in (-\infty, \infty)$ and $F_0 \in \mathcal{F}_0$ is the class of all symmetric distributions (around 0). $\square$

### Parametric or nonparametric inference ?

Let us assume that our objective is to estimate $\theta := F(s) = P(X \leq s) \in [0, 1]$ for a given $s$.

- Without making any assumption about the distribution of $X$, a reasonable estimator is the empirical cdf $\hat{F}(s) = n^{-1} \sum_i I(X_i \leq s)$.
- If we assume that $X \sim N(\mu, 1)$, then it is more reasonable to estimate $\theta$ by $\Phi(s - \hat{\mu})$, where $\Phi$ is the cdf of a $N(0, 1)$ and $\hat{\mu}$ is any given estimator of $\mu$.

  $\rightarrow$ Incorrect assumptions and restrictions on the underlying distribution of $X$ will lead to wrong or ***biased*** conclusions. But correct assumptions lead to better and more ***efficient*** estimation.

## 1.3 Identifiability

Identifiability is an important property of a statistical model, which determines whether the parameters of the model can be recovered from the observed data. An identifiability problem arises when different parameter values can rise to identical probability distributions and therefore identical samples. Why is this a problem?
$\rightarrow$ the data can tell us about $\theta$ if only different values of $\theta$ lead to different samples.

**Definition 1.2.** A parameter of interest is statistically identifiable if it can be *uniquely determined* from the distribution of the *observed sample*.

Mathematically, this can be formulated by saying that $\theta \mapsto P_\theta$ is one-to-one, i.e.

$$P_{\theta_1} = P_{\theta_2} \Rightarrow \theta_1 = \theta_2, \ \forall \theta_1, \theta_2 \in \Theta.$$

When a model is not identifiable, we can find $\theta_1 \neq \theta_2$ such that $P_{\theta_1} = P_{\theta_2}$. In this case, we cannot distinguish these two values of $\theta$ by examining a sample from $P_\theta$, even if its size is huge/infinite.

**Example 1.3.**

- The Bernoulli, the Exponential, and the Normal models, as defined above, are identifiable.
- If we observe $X = \mu_1 + \epsilon$, where $\epsilon \sim N(\mu_2, 1)$ and $\mu_1$ and $\mu_2$ are unknown, then $\theta = \mu_1 + \mu_2$ is identifiable but $\boldsymbol{\theta} = (\mu_1, \mu_2)^t$ is not. $\square$

**Exercise 1.1.** Suppose that we observe $X = |Y|$, with $Y \sim N(\mu, 1)$. Show that $\mu$ is not identifiable.

If a model is not identifiable, it is common to introduce additional constraints/assumptions on it in order to make it identifiable. In that case, the set of these requirements is called the ***identifiability conditions.***

## 1.4 Misspecification

If the true but unknown distribution the observed data ($X$) is not a member of the assumed family $\mathcal{P}$ then we say that the model is misspecified. More precisely, a parametric model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is correctly specified if $\exists \theta_0 \in \Theta$ such that $X \sim P_{\theta_0}$. In this case, we refer to $\theta_0$ as the ***true value*** of the parameter $\theta$.

*Attention*: For notational simplicity, when no confusion is possible, we leave off the subscript 0 in $\theta_0$ and use $\theta$ to denote both the unknown generic parameter of interest and its true value.

An example of misspecification is when we fit a Normal distribution to observations which come from, for example, an Exponential distribution.

A conclusion drawn from a statistical model is ***valid only if the chosen model is correctly specified***. Realistically, the performance of a model depends on how closely it approximates the true distribution of the observed data.

Essentially, all models are wrong, but some are useful— George Box.

From now one, unless mentioned explicitly, we assume that any model we are fitting to the data is correctly specified.

## 1.5 Purpose of inferential statistics

**Definition 1.3.** Statistical inference is the process of learning about probability models using observed data.

To be more specific, suppose we have observed data $\boldsymbol{x}$ according to a parametric model $\mathcal{P} = \{P_\theta, \boldsymbol{\theta} \in \Theta\}$. The goal of parametric statistical inference is then to learn about $\boldsymbol{\theta}$ from $\boldsymbol{x}$.

There are three major parametric statistical inference procedures:

1. **Point estimation**: A single value is computed from the data $\boldsymbol{x}$ and used as an estimate of the parameter of interest $\boldsymbol{\theta}$.
2. **Hypothesis testing**: Sets up some specific hypotheses regarding $\boldsymbol{\theta}$ and assesses whether or not the data $\boldsymbol{x}$ support these hypothesis.
3. **Confidence set estimation**: Use the observed data $\boldsymbol{x}$ to construct a set of values for $\boldsymbol{\theta}$. The resulting set must have a high (predetermined) probability of including the true value.

Other well-known topics in statistical inference include model selection, model validation and prediction.

# 2 Exponential family

An exponential family is a practically convenient and widely used united family of distributions. It contains most of the standard discrete and continuous distributions that are used for modeling, such as (multivariate) normal, poisson, binomial, multinomial, exponential, and gamma.

The reason for the special status of exponential family is that a number of important and useful results in inference can be unified within the framework of exponential family. This family also forms the basis for an important class of regression models, known as generalized linear models.

## 2.1 One-parameter exponential family

A family of probability distributions that depend on a single parameter $\theta$ is a one-parameter exponential family if it can be expressed as

$$f(x;\theta) = h(x)\exp(g(\theta)T(x) - B(\theta)),\ \forall x.$$

Here $x$ can be a scalar or vector, $h(x) \geq 0$ and $T(x)$ are a functions of $x$ only (*cannot depend on* $\theta$), and $g(\theta)$ and $B(\theta)$ are functions of $\theta$ only (*cannot depend on* $x$). $B(\theta)$ is a normalizing constant, ensuring that $f(x;\theta)$ sums or integrates to 1. $T(X)$ is referred to as ***natural sufficient*** statistic or simply natural statistic.

Note that the above parameterization is not unique since, for example, $g$ could be multiplied by a nonzero constant $a$ if $T$ is divided by $a$. Also, in many cases $T(x) = x$.

An exponential family can be reparameterized as

$$h(x)\exp(\eta T(x) - A(\eta)).$$

This expression is called the ***canonical (or natural) representation***, and $\eta = g(\theta)$ is the ***canonical parameter***. Here $h(x)$ and $T(x)$ are the same as in the original parameterization, and $A(\eta) := B\big(g^{-1}(\eta)\big)$ is the new normalizing constant (assuming $g$ is invertible). Notice that this parametrization is not unique either.

It's analytically convenient and easier to work with an exponential family in its canonical form. Once a result has been derived for the canonical form, we can rewrite it in terms of the original parameter $\theta$ if desired.

To verify that a family of pd's is an exponential family, we must identify all the functions $h$, $T$, $g$, and $B$ (or $A$). The next example illustrates this.

**Example 2.1.**

- Poisson:

$$\begin{aligned}\frac{\theta^x e^{-\theta}}{x!} &= \frac{1}{x!}\exp(x\log(\theta)-\theta),\ \ x=0,1,\ldots,\ \text{and}\ \theta>0\\ &\equiv \frac{1}{x!}\exp(x\eta-e^{\eta}),\ \ \eta\in(-\infty,\infty).\end{aligned}$$

- Binomial:

$$\begin{aligned}C_n^x\theta^x(1-\theta)^{n-x} &= C_n^x\exp\left(x\log\left(\frac{\theta}{1-\theta}\right)+n\log(1-\theta)\right),\ \ x=0,1,\ldots,n\ \text{and}\ \theta\in(0,1)\\ &\equiv C_n^x\exp(x\eta-n\log(1+e^{\eta})),\ \ \eta\in(-\infty,\infty).\end{aligned}$$

- Normal with a *known* $\sigma$ ($\theta=\mu$):

$$\begin{aligned}\frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) &= \frac{1}{\sqrt{2\pi}\sigma}\exp\left(\frac{-x^2}{2\sigma^2}\right)\exp\left(\frac{\mu}{\sigma^2}x-\frac{\mu^2}{2\sigma^2}\right),\ \ x\in(-\infty,\infty)\ \text{and}\ \mu\in(-\infty,\infty)\\ &\equiv \frac{1}{\sqrt{2\pi}\sigma}\exp\left(\frac{-x^2}{2\sigma^2}\right)\exp\left(\eta x-\frac{\eta^2\sigma^2}{2}\right),\ \ \eta\in(-\infty,\infty).\ \square\end{aligned}$$

**Example 2.2** (Counter-example)**.** The following density

$$\begin{aligned} f(x;\theta) &= \exp(-(x-\theta)),\ \ x \geq \theta \text{ and } \theta \in (-\infty, \infty) \\ &= \exp(-(x-\theta))I(x \geq \theta) \end{aligned}$$

is not an exponential family because of $I(x \geq \theta)$ that cannot be factored into "$x$" part and an "$\theta$" part. □

In general, for a pd $f(x;\theta)$ to be an exponential family, its *support* $S = \{x : \ f(x;\theta) > 0\}$ *must be free from* $\theta$.

## 2.2 Properties of exponential family

A special fact about the exponential family is that the integral $\int h(x)\exp(g(\theta)T(x) - B(\theta))dx$, in the continuous case, or the sum $\sum_x h(x)\exp(g(\theta)T(x) - B(\theta))$, in the discrete case, can be differentiated, with respect to $\theta$ (or with respect $\eta$ for the canonical representation), any number of times under the integral/sum sign (i.e. the derivatives can always be moved inside the integral/sum). This is the source of many interesting results about the exponential family. One of these results is given in the following.

**Proposition 2.1.** *With the canonical parameterization, the mean and variance of $T \equiv T(X)$ are given by*

$$E_\eta(T) = A^{'}(\eta) \text{ and } Var_\eta(T) = A^{''}(\eta).$$

With the original parameterization, we can write

$$E_\theta(T) = \frac{B^{'}(\theta)}{g^{'}(\theta)} \text{ and } Var_\theta(T) = \frac{(E_\theta(T))^{'}}{g^{'}(\theta))} = \frac{g^{'}(\theta)B^{''}(\theta) - g^{''}(\theta)B^{'}(\theta)}{(g^{'}(\theta))^3}.$$

For example, for $N(\mu, \sigma^2)$, given the results above (see Example 2.1), we see that

$$E(X) = \partial_\eta\left(\frac{\eta^2\sigma^2}{2}\right) = \eta\sigma^2 = \mu \quad \text{and} \quad Var(X) = \partial_\eta(\eta\sigma^2) = \sigma^2$$

Another interesting fact about the exponential family is that its structure is preserved under iid sampling. This is better explained in the following.

**Proposition 2.2.** *If $X_1, \ldots, X_n$ are iid rv from the exponential family, as defined above, with a natural statistic $T$ then the the*

*joint distribution of* $\boldsymbol{X} = (X_1, \ldots, X_n)$*:*

$$\left[\prod_{i=1}^{n} h(x_i)\right] \exp\left(\eta \sum_{i=1}^{n} T(x_i) - nA(\eta)\right)$$

*is also an exponential family with natural statistic* $\sum_{i=1}^{n} T(X_i)$*.*

## 2.3 Multiparameter exponential family

The multiparameter version of the exponential family is given by

$$\begin{aligned} f(x;\boldsymbol{\theta}) &= h(x) \exp\left(\sum_{j=1}^{J} g_j(\boldsymbol{\theta}) T_j(x) - B(\boldsymbol{\theta})\right) \\ &\equiv h(x) \exp\big(\boldsymbol{\eta}^t \boldsymbol{T}(x) - A(\boldsymbol{\eta})\big), \qquad \text{(Canonical parametrization)} \end{aligned}$$

where $\boldsymbol{\theta}^t = (\theta_1, \ldots, \theta_J)$, $\boldsymbol{\eta}^t = (\eta_1, \ldots, \eta_J)$, $\boldsymbol{T}^t(x) = (T_1(x), \ldots, T_J(x))$, and $g_j, A, B : \mathbb{R}^J \to \mathbb{R}$.

The properties that we have seen above for the one-parameter case also apply to the multi-parameter case. For

example, it can be shown that

$$E(T_j(X)) = \partial_{\eta_j} A(\boldsymbol{\eta}),$$
$$Cov(T_j(X), T_k(X)) = \partial_{\eta_j \eta_k} A(\boldsymbol{\eta}).$$

In matrix form, we can write $E(\boldsymbol{T}(X)) = \boldsymbol{\nabla} A(\boldsymbol{\eta})$ and $Var(\boldsymbol{T}(X)) = \boldsymbol{\nabla}^2 A(\boldsymbol{\eta})$.

**Example 2.3.** Normal distribution with an unknown $\mu$ and $\sigma$ ($\boldsymbol{\theta} = (\mu, \sigma^2)$):

$$\begin{aligned}\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) &= \frac{1}{\sqrt{2\pi}} \exp\left(\frac{\mu}{\sigma^2}x - \frac{1}{2\sigma^2}x^2 - \left(\frac{\mu^2}{2\sigma^2} + \log(\sigma)\right)\right) \\ &\equiv \frac{1}{\sqrt{2\pi}} \exp\left(\eta_1 x + \eta_2 x^2 - A(\boldsymbol{\eta})\right),\end{aligned}$$

where $A(\boldsymbol{\eta}) = -\frac{\eta_1^2}{4\eta_2} - \frac{1}{2}\log(-\eta_2) - \log\left(\sqrt{2}\right)$. Here $T_1(x) = x$ and $T_2(x) = x^2$.

It follows that, $\partial\eta_1 A(\boldsymbol{\eta}) = -\frac{\eta_1}{2\eta_2} = \mu = E(X)$, $\partial\eta_2 A(\boldsymbol{\eta}) = \frac{\eta_1^2 - 2\eta_2}{4\eta_2^2} = \mu^2 + \sigma^2 = E(X^2)$, $\partial^2\eta_1 A(\boldsymbol{\eta}) = -\frac{1}{2\eta_2} = \sigma^2 = Var(X)$, $\partial\eta_1\eta_2 A(\boldsymbol{\eta}) = \frac{\eta_1}{2\eta_2^2} = 2\mu\sigma^2 = Cov(X, X^2)$, and $\partial^2\eta_2 A(\boldsymbol{\eta}) = \frac{\eta_2 - \eta_1^2}{2\eta_2^3} = 2\sigma^2(\sigma^2 + 2\mu^2) = Var(X^2)$. $\square$

# 3 Some useful tools

### Law of total expectation/variance

For any two rv $X$ and $Y$,

$$E(Y) = E\left(E[Y|X]\right),$$
$$Var(Y) = E\left(Var[Y|X]\right) + Var\left(E[Y|X]\right).$$

### Expected value of a non-negative rv

If $X$ be a non-negative rv, then $E(X) \geq 0$. Moreover, $E(X) = 0$ if and only if $X = 0$ (with probability 1).

### Markov-Chebyshev's inequality

*If $X$ is a non-negative rv, then $E(X) \geq kP(X \geq k)$, $\forall k \in \mathbb{R}$.* As a consequence, for any rv $X$ and any constant $k > 0$, $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$, where $\mu = E(X)$ and $\sigma = \sqrt{Var(X)}$.

### Jensen's inequality

If $g$ is a convex function ($g'' \geq 0$), then

$$g(E(X)) \leq E(g(X))$$

Moreover, if $g$ is strictly convex, then this inequality is strict unless $X$ is constant.
The opposite hold for concave function ($g$ is concave if and only if $-g$ is convex).

**Example 3.1.** $|E(X)| < E|X|$, and $(E(X))^2 < E(X^2)$. If $X > 0$, $E\left(\sqrt{X}\right) < \sqrt{E(X)}$, and $E(\log(X)) < \log(E(X))$. $\square$

### Cauchy-Schwarz's inequality

For any two rv $X$ and $Y$,

$$(E(XY))^2 \leq E(X^2)E(Y^2)$$

As a consequence, we get the inequality

$$(Cov(X,Y))^2 \leq Var(X)Var(Y).$$

Moreover, if $X$ and $Y$ are not constant, then the last inequality becomes an equality if and only if $X$ and $Y$ are linearly dependent, i.e. if and only if there exists numbers $a \neq 0$ and $b$ such that $Y = aX + b$, with probability one.

### Taylor's theorem

If a function $f$ is $k \geq 1$ times differentiable at a given point $a$, then

$$f(x) = f(a) + \sum_{i=1}^{k} \frac{1}{i!}(x-a)^i f^{(i)}(a) + (x-a)^k r_a(x),$$

for some function $r_a$ with $\lim_{x \to a} r_a(x) = 0$. This $r_a$ is called the remainder. Typically, we will not be concerned with the explicit form of this remainder. Since we are interested in approximations, we are just going to ignore the remainder and simply write

$$f(x) \approx f(a) + \sum_{i=1}^{k} \frac{1}{i!}(x-a)^i f^{(i)}(a).$$

### Some interesting properties of the Variance-Covariance matrix

Let $\boldsymbol{X}$ be a rve in $\mathbb{R}^d$ and $\boldsymbol{\Sigma} = Var(\boldsymbol{X}) = \left(Cov(X_j, X_k)\right)_{1 \leq j,k \leq d}$. $\boldsymbol{\Sigma}$ is symmetric ($\boldsymbol{\Sigma}^t = \boldsymbol{\Sigma}$) and positive semidefinite ($\boldsymbol{a}^t \boldsymbol{\Sigma} \boldsymbol{a} \geq 0$, $\forall \boldsymbol{a} \in \mathbb{R}^d$; we write $\boldsymbol{\Sigma} \succeq 0$). This last property follows directly from the fact that $\boldsymbol{a}^t \boldsymbol{\Sigma} \boldsymbol{a} = Var(\boldsymbol{a}^t \boldsymbol{X})$.

$\boldsymbol{\Sigma}$ is positive definite ($\boldsymbol{a}^t \boldsymbol{\Sigma} \boldsymbol{a} > 0$, $\forall \boldsymbol{a} \neq \boldsymbol{0}$; we write $\boldsymbol{\Sigma} \succ 0$) if and only if the components of $\boldsymbol{X}$ are linearly independent (with probability 1); i.e. it is not possible to express any component of $\boldsymbol{X}$ (i.e. $X_1$ or $X_2$, ...) as a linear combination of the others.

A positive definite matrix is invertible and its inverse is also positive definite.

For a positive semidefinite (definite) matrix $\boldsymbol{A}$, there exists a unique positive semidefinite (definite) matrix $\boldsymbol{B}$ such that $\boldsymbol{B} \times \boldsymbol{B} = \boldsymbol{A}$. This $\boldsymbol{B}$ is called the square-root of $\boldsymbol{A}$ and is denoted by $\boldsymbol{A}^{1/2}$.

### Some properties of the multivariate normal

- Let $\boldsymbol{X} \sim N_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{B}$ a $p \times d$ matrix and $\boldsymbol{a}$ $p-$dimensional vector, then

$$\boldsymbol{a} + \boldsymbol{B}\boldsymbol{X} \sim N_p\left(\boldsymbol{a} + \boldsymbol{B}\boldsymbol{\mu}, \boldsymbol{B}\boldsymbol{\Sigma}\boldsymbol{B}^t\right).$$

As a corollary, if $\boldsymbol{b}$ $d-$dimensional vector, then $\boldsymbol{b}^t\boldsymbol{X} \sim N\left(\boldsymbol{b}^t\boldsymbol{\mu}, \boldsymbol{b}^t\boldsymbol{\Sigma}\boldsymbol{b}\right)$.

- $\boldsymbol{X} \sim N_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \Rightarrow \boldsymbol{\Sigma}^{-1/2}(\boldsymbol{X} - \boldsymbol{\mu}) \sim N_d(\mathbf{0}, \mathbb{1})$, where $\boldsymbol{\Sigma}^{-1/2}$ is the square-root of a of $\boldsymbol{\Sigma}^{-1}$, $\mathbb{1}$ is the identity matrix, and $\mathbf{0}$ the $d-$dimensional vector of zeros.
- $\boldsymbol{X} \sim N_d(\mathbf{0}, \mathbb{1}) \Rightarrow \boldsymbol{X}^t\boldsymbol{X} \sim \chi^2_d$, where $\chi^2_d$ the chi-squared distribution with $d$ degrees of freedom.